Understanding Compensation Inequality in NYC Payroll Data: An Exploratory Analysis of Salary, Overtime, and High-Pay Status

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Introduction

Public payroll data is useful because it shows more than just how much employees are paid. It also reveals how compensation is structured across different agencies, work locations, and job roles. In a large city such as New York, employee income may come from several sources, including base salary, overtime pay, and other forms of compensation. Because of this, total compensation may differ a lot even among workers in the same public system. Studying payroll data can therefore help us better understand income inequality, overtime dependence, and the role of organizational structure in shaping employee earnings.

This project conducts an exploratory data analysis of New York City employee payroll data. The dataset contains about 10,000 employee records and 17 variables, including both numeric and categorical features such as agency name, work borough, job title, pay basis, base salary, regular hours, overtime hours, overtime pay, and other pay. The project focuses on three main research questions. First, how is total compensation distributed across different income components? Second, how do compensation patterns vary across agencies, boroughs, job titles, and pay basis types? Third, can high-paying employees be identified using observable employee characteristics? These questions allow the project to move from simple description to deeper interpretation of how compensation is structured in the public workforce.

The overall argument of this report is that NYC employee compensation is not determined by base salary alone. Instead, total earnings are shaped by a layered structure that includes regular pay, overtime participation, and institutional differences across agencies and occupations. To support this argument, the project combines exploratory

data analysis with a lightweight machine learning exercise. The machine learning models are not used to make causal claims, but rather to test whether high-pay status follows recognizable patterns in the data.

Dataset and Research Design

The dataset used in this project comes from New York City employee payroll records accessed through a public data source. The original sample includes 10,000 observations and 17 variables. These variables include employee characteristics, compensation components, and job-related information. Important variables include fiscal_year, agency_name, work_location_borough, title_description, pay_basis, base_salary, regular_hours, regular_gross_paid, ot_hours, total_ot_paid, total_other_pay, and agency_start_date. Because the assignment required mixed variable types and at least 300 rows, this dataset clearly satisfies the technical requirements of the project.

The project is designed as an exploratory analysis rather than a causal study. This means the goal is not to prove that one variable directly causes another. Instead, the goal is to identify meaningful patterns, describe compensation structure, compare groups, and test whether certain features are associated with higher pay. This is appropriate for payroll data because the dataset contains rich descriptive information, but it does not include all the background factors needed for strong causal inference. For example, the data cannot fully explain promotions, performance, or agency-specific staffing rules.

To answer the research questions, the project uses two stages of analysis. The first stage is exploratory data analysis, which examines distributions, group differences, and relationships among variables. The second stage is a simple prediction task in which employees are classified into a high-pay group or a lower-pay group. Logistic Regression and Random Forest are used for this purpose. This design allows the project

to combine interpretation and validation: EDA shows the patterns, while the models test whether those patterns are stable enough to predict high-pay status.

Data Cleaning and Feature Engineering

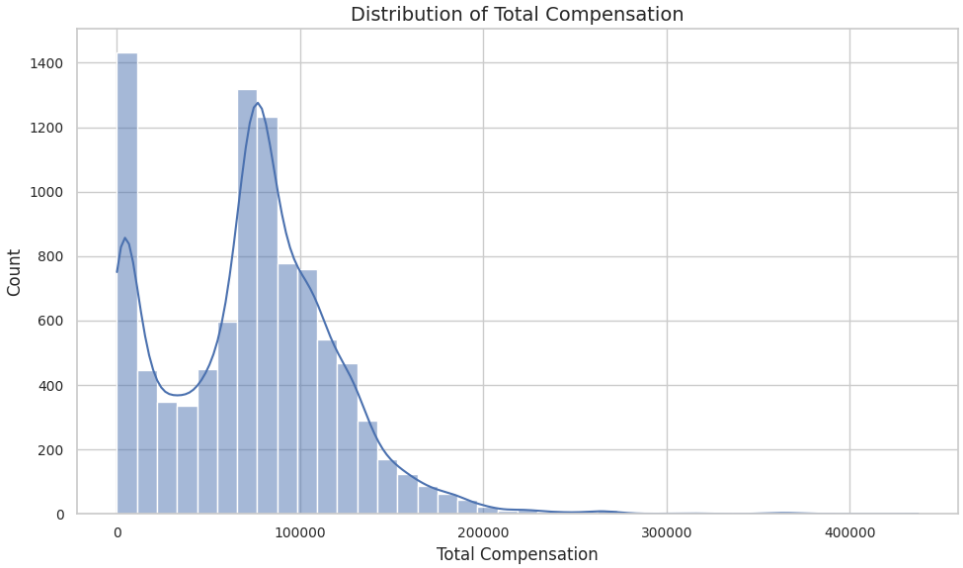
Before the main analysis, the dataset required cleaning and transformation. Many variables were initially stored as strings even though they represented numeric values. As a result, salary, hours worked, overtime pay, and other compensation fields were converted to numeric format. In addition, agency_start_date was converted from string form into a proper datetime variable. These steps were necessary so that the numeric variables could be analyzed correctly and so that new time-based features could be created.

The project also handled missing values and removed abnormal records. In particular, observations with negative work hours, negative compensation amounts, or negative tenure were treated as invalid for the purposes of this analysis. The cleaned dataset was then used for both EDA and machine learning. This cleaning step matters because payroll records sometimes include administrative corrections that can distort ordinary patterns. Removing abnormal values made distributions and comparisons easier to interpret.

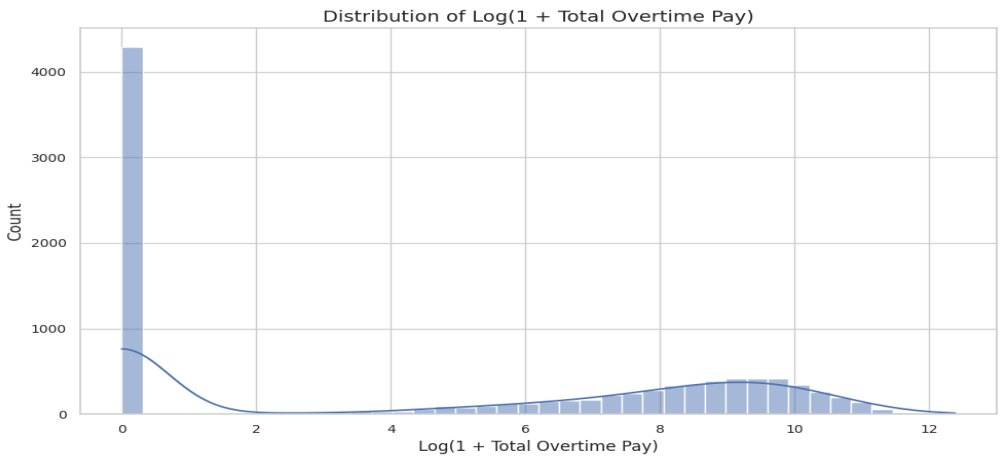
Several new variables were engineered to better answer the research questions. First, total_compensation was created by summing regular gross pay, overtime pay, and other pay. Second, tenure_years were calculated based on the employee's agency start date. Third, overtime_share and other_pay_share were created to measure the relative contribution of different compensation components. Finally, a binary target variable called high_pay was created, where employees above the median total compensation were classified as 1 and others as 0. These engineered variables were especially useful because they translated raw payroll information into interpretable measures of compensation structure and employee status.

Exploratory Data Analysis

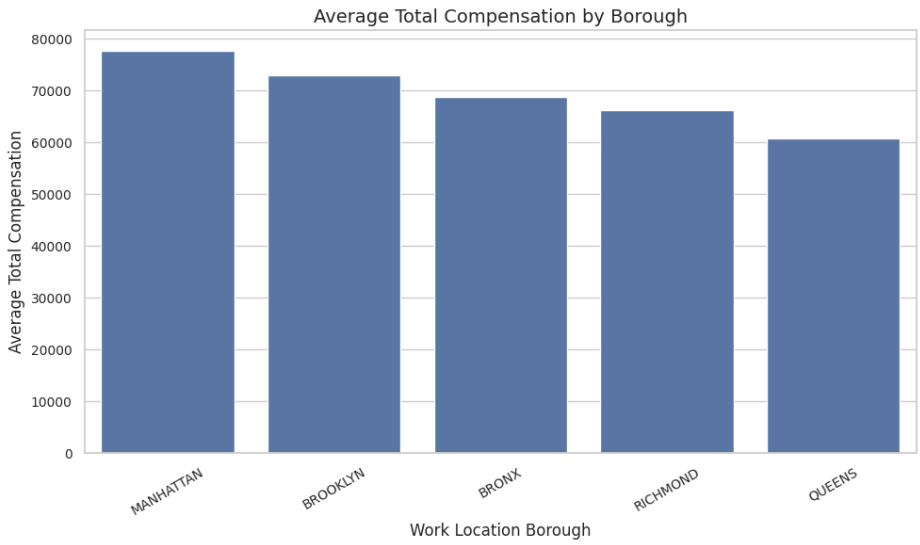
The first major finding from the EDA is that compensation is strongly right skewed. The distribution of total compensation shows that most employees earn within a moderate range, while a relatively small number of employees receive much higher compensation. In contrast, base salary is more concentrated. This suggests that fixed salary alone does not fully explain why some employees earn much more than others. Instead, the wider spread in total compensation appears to come from non-salary components, especially overtime pay and other forms of additional income.



A second important finding is that overtime pay is highly unevenly distributed. Even after log transformation, overtime remains strongly skewed, with many employees receiving little or no overtime and a smaller group receiving very large amounts. This pattern suggests that overtime is not a universal supplement to salary. Rather, it acts as a major differentiator for certain employees and roles. In practical terms, this means that compensation inequality is not only about salary differences; it is also about whether an employee is positioned to receive large overtime payments.



The grouped analysis also shows strong heterogeneity across boroughs and agencies. Employee counts are not evenly distributed across work locations, with Manhattan having the largest concentration of employees. Average compensation also differs across boroughs. Average compensation also differs across boroughs, with Manhattan appearing to have the highest average total compensation among major groups. However, this difference likely reflects variation in job composition, agency concentration, and occupational structure across boroughs, rather than purely geographic effects.. At the same time, compensation varies significantly across agencies. Some agencies consistently pay more on average, which may reflect different operational demands, job structures, skill requirements, or overtime opportunities. These patterns suggest that compensation is shaped partly by organizational context rather than only by individual characteristics.



Job title analysis supports the same conclusion. Certain titles appear repeatedly among the highest-paying roles, especially positions linked to supervision, field operations, legal work, or social service management. This indicates that high compensation is associated not just with generic employment in city government, but with specific occupational categories. Different job roles likely involve different responsibilities, workloads, and opportunities for additional pay. Therefore, compensation inequality in the dataset should be understood as structural and role-based rather than purely random.

The compensation composition analysis provides a useful final piece of interpretation. Regular pay remains the largest component of total earnings for most workers, which means base salary still forms the foundation of compensation. However, overtime represents a meaningful additional share for some employees, especially those in the upper tail of income distribution. This creates at least three broad compensation patterns: salary-dominant employees, overtime-heavy employees, and employees with a more mixed compensation structure. This finding is important because it shows that the public workforce is not financially homogeneous. Employees may belong to the same overall payroll system while still following very different earning patterns.

Overall, the EDA supports a clear interpretation: total compensation in NYC payroll data is layered. Base salary provides the baseline, but overtime and institutional differences create much of the variation that separates higher earners from the rest. The distribution of compensation is therefore unequal not only because of salary ranking, but also because of how work hours, overtime access, and organizational placement interact.

Predictive Analysis

To extend the descriptive analysis, the project uses machine learning to predict whether an employee belongs to the high-pay group. Importantly, this step is designed as a pattern recognition exercise rather than a causal analysis. The goal is to evaluate whether high-pay status can be reliably identified using observable employee and job-related characteristics.

The target variable, high_pay, is defined as whether an employee's total compensation exceeds the median. Features used in the models include base salary, regular hours, tenure, agency, job title, borough, pay basis, and leave status. Categorical variables with high cardinality, such as agency and title, were simplified before applying one-hot encoding. Numerical variables were standardized to ensure comparability across features.

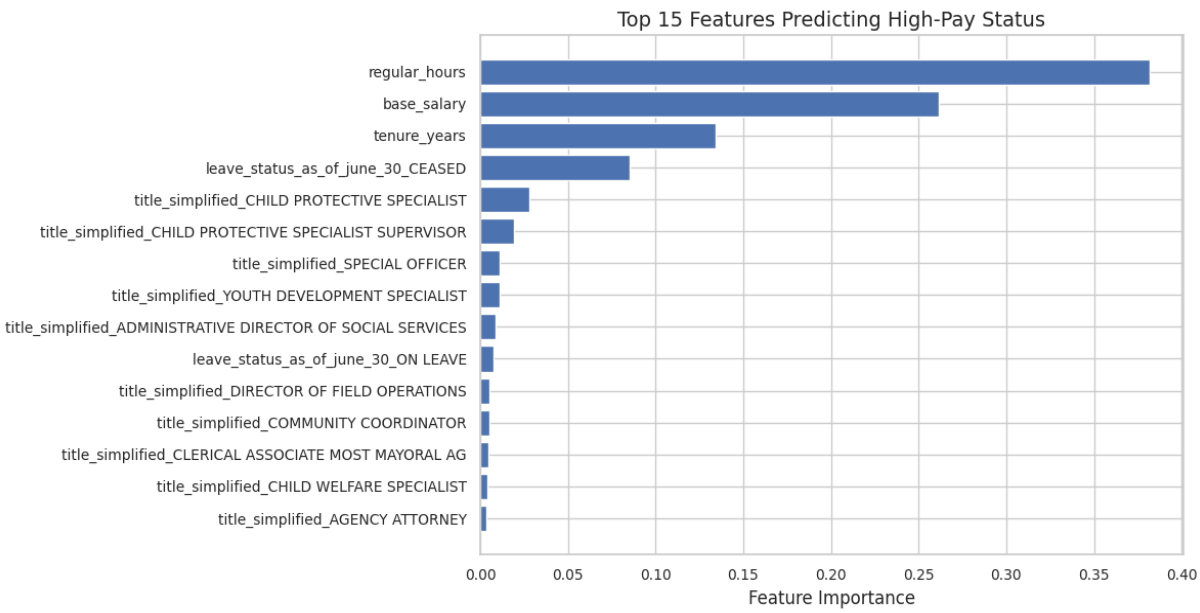
Two models were implemented: Logistic Regression with cross-validation and Random Forest. Both models achieved strong performance, with accuracy around 90% and ROC-AUC values close to 0.96. A 5-fold cross-validation procedure produced consistent results, indicating that the models are stable and not driven by random variation in the training data.

It is important to interpret these results with caution. Base salary is a direct component of total compensation, which means that part of the predictive performance reflects built-in relationships within the data. As a result, the models should be understood as identifying patterns in compensation structure rather than making fully

independent predictions. The high performance therefore indicates that high-pay status is strongly associated with observable compensation-related features.

Feature importance analysis from the Random Forest model shows that base salary, regular hours, and tenure are among the most influential predictors. In addition, certain job titles and employment conditions contribute meaningfully to prediction. These findings are consistent with the exploratory analysis, suggesting that higher compensation is associated with workload, formal pay level, and organizational role. However, it is important to emphasize that feature importance reflects statistical association rather than causal relationships.

Overall, the machine learning results reinforce the main conclusion of the project: high compensation in NYC payroll data follows a structured and predictable pattern based on job characteristics and compensation components, rather than arising randomly.



Limitations and Implications

Although the findings are meaningful, the project has several limitations. First, the dataset is a sample rather than the full population of all NYC employees. Second, payroll data may contain administrative irregularities, missing values, or special cases that are difficult to interpret. Third, the project is observational, so it cannot establish causality. For example, even if overtime is associated with high compensation, this does not prove that overtime alone causes higher earnings in a broader policy sense. There may be unobserved factors such as promotion history, department rules, or unique job assignments.

Even with these limitations, the analysis has useful implications. If a small number of employees depend heavily on overtime, this may point to staffing imbalances, scheduling pressure, or uneven workload distribution. Large agency-level differences may also raise questions about workforce structure and pay equity. More broadly, the project shows that payroll data can be used not only to describe compensation, but also to identify patterns relevant to management and policy decisions. However, such interpretations should remain cautious and descriptive unless stronger causal data becomes available.

Conclusion

This project used NYC payroll data to examine compensation structure, group differences, and high-pay prediction. The results show that employee earnings are highly uneven and that total compensation is shaped by more than base salary alone. Overtime plays a major role for some employees, while agency, role, and workload create additional layers of inequality. The machine learning results support the same general conclusion by showing that high-pay status can be predicted fairly accurately using a limited set of job-related features. Overall, the analysis suggests that NYC employee compensation reflects a structured system of salary, overtime participation, and institutional differences rather than a simple single-pay framework.

References

New York City Open Data. Citywide Payroll Data.

Course project notebook prepared by the authors.